Lec-31

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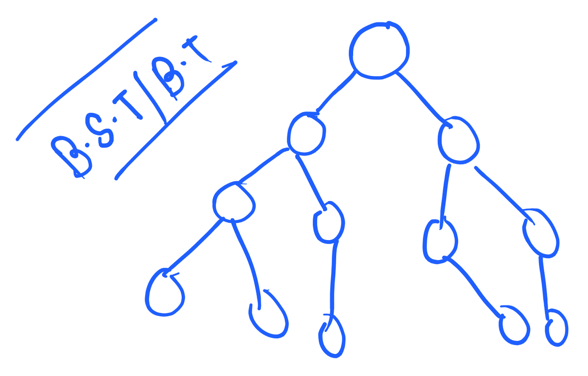
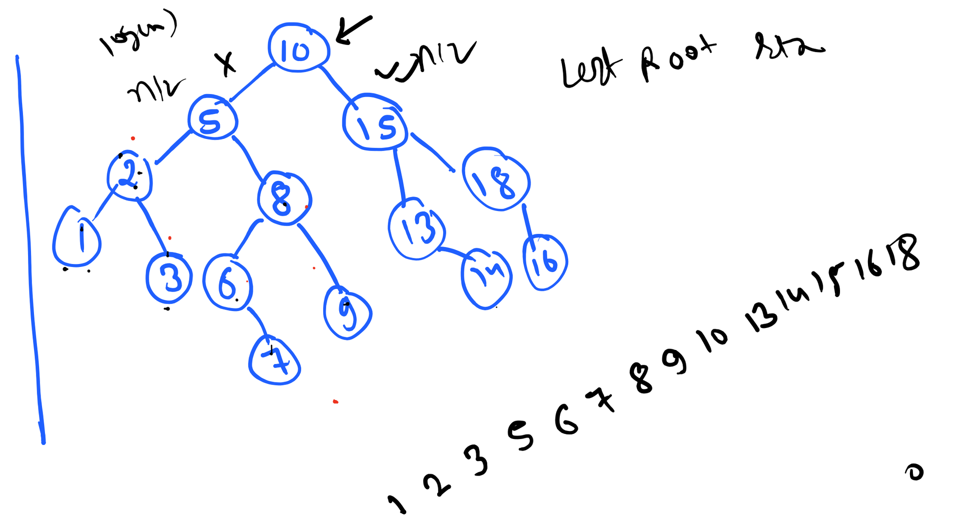
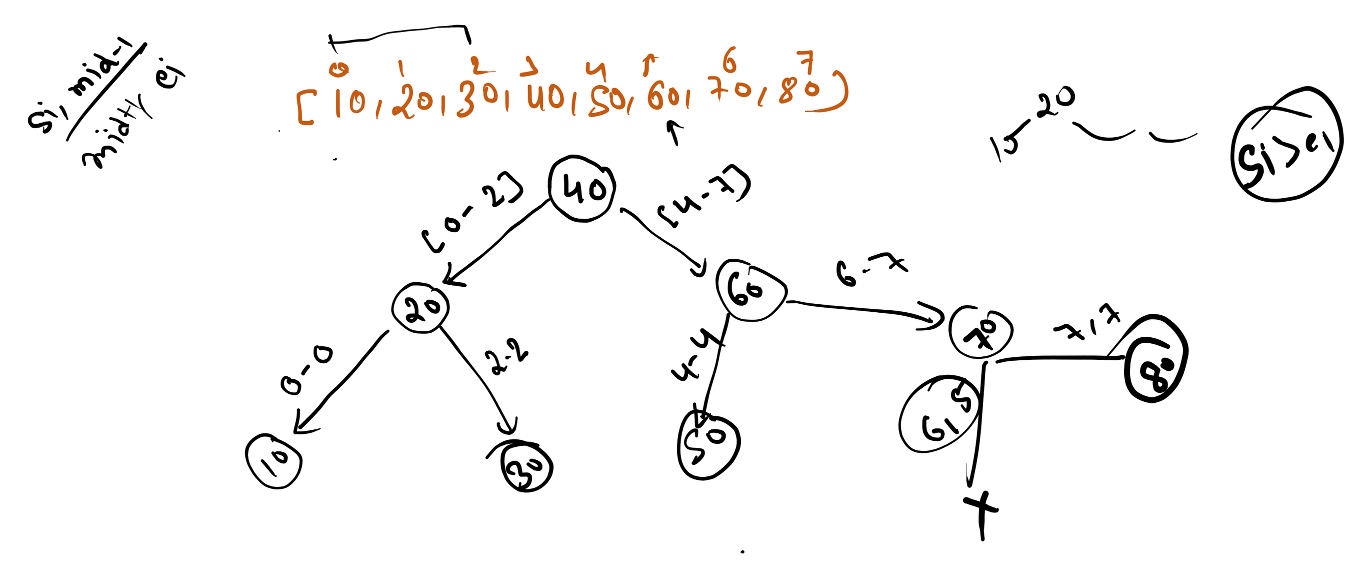
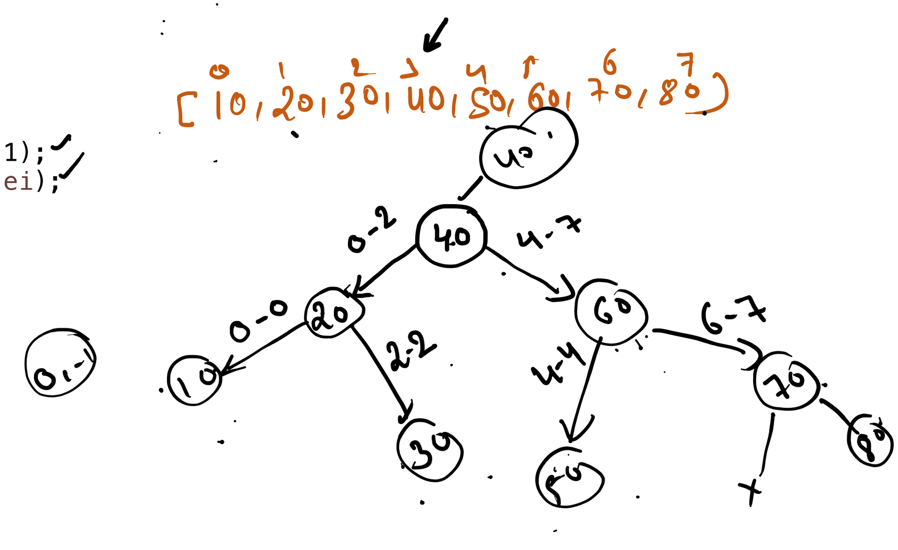






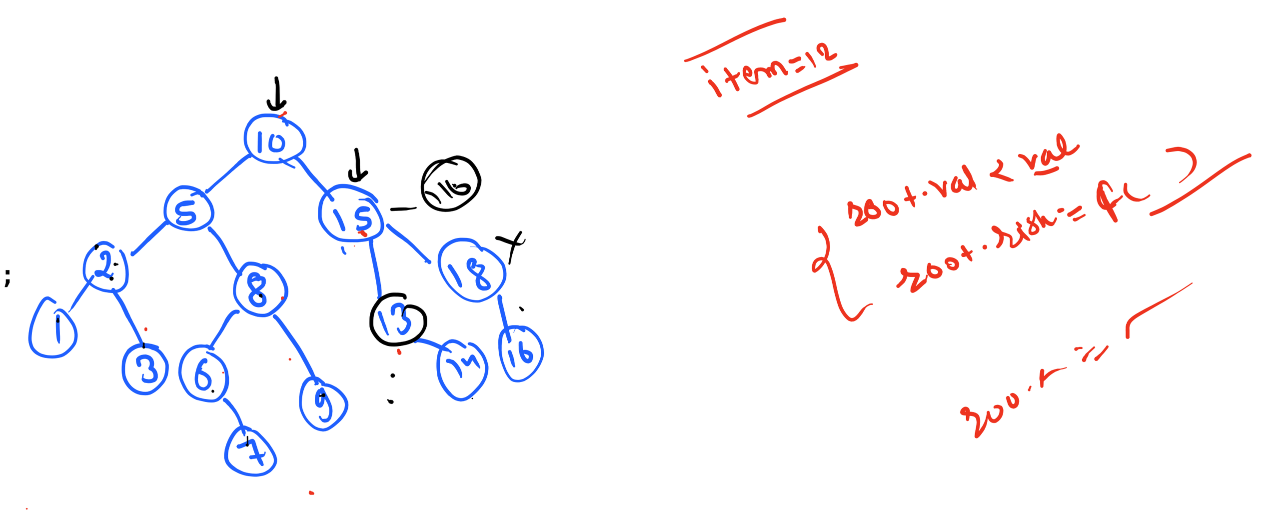

```java
private Node Create(int[] in, int si, int ei) {
        // TODO Auto-generated method stub
        if (si>ei) {
            return null;
        }
        int mid = (si + ei) / 2;
        Node nn = new Node(in[mid]);
        nn.left = Create(in, si, mid - 1);
        nn.right = Create(in, mid + 1, ei);
        return nn;
    }
```




```java
public TreeNode insertIntoBST(TreeNode root, int val) {

        if (root == null) {
            return new TreeNode(val);
        }
        if (root.val < val) {
            root.right = insertIntoBST(root.right, val);

        } else {
            root.left = insertIntoBST(root.left, val);
        }
        return root;
    }
```



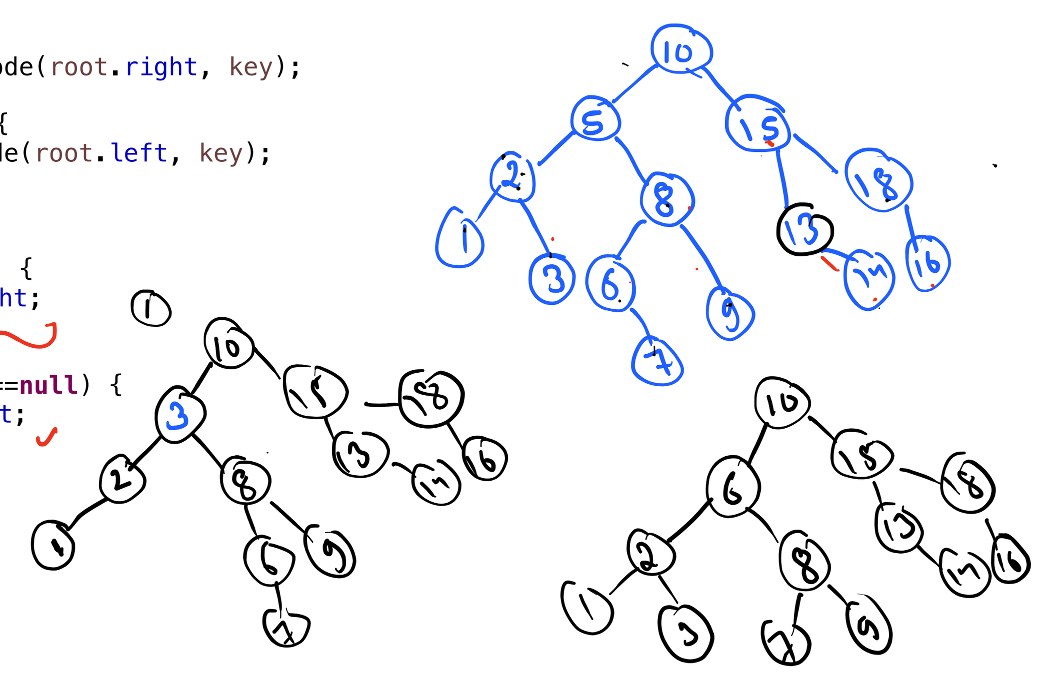

```java
public TreeNode deleteNode(TreeNode root, int key) {
        if(root==null) {
            return null;
        }
        if(root.val<key) {
            root.right=deleteNode(root.right, key);
        }
        else if(root.val>key) {
            root.left=deleteNode(root.left, key);
        }
        else {
            // 1 child
            if(root.left==null) {
                return root.right;

            }
            else if(root.right==null) {
                return root.left;

            }

        }

    }
```

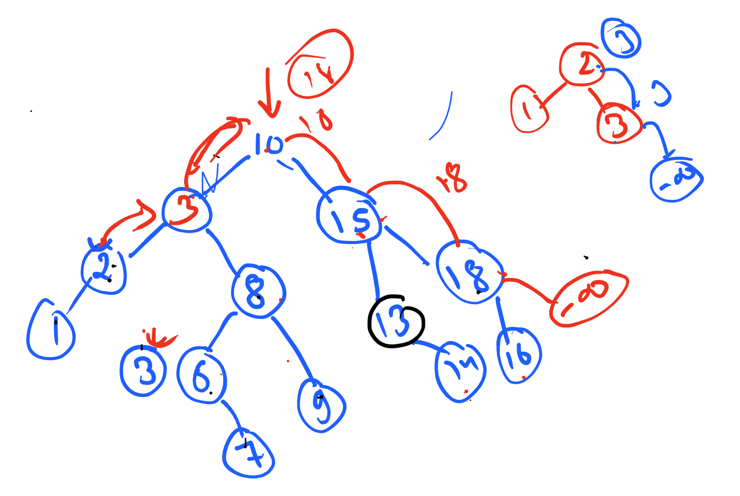


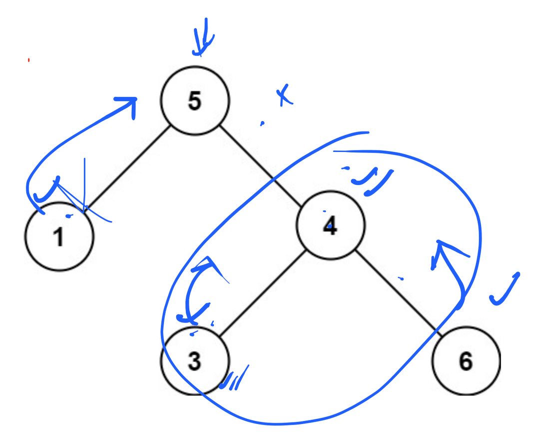
```java
public TreeNode deleteNode(TreeNode root, int key) {
        if (root == null) {
            return null;
        }
        if (root.val < key) {
            root.right = deleteNode(root.right, key);
        } else if (root.val > key) {
            root.left = deleteNode(root.left, key);
        } else {
            // 1 or 0 child
            if (root.left == null) {
                return root.right;

            } else if (root.right == null) {
                return root.left;
            } else {
                int max = max(root.left); →3
                root.left = deleteNode(root.left, max);
                root.val = max;
            }

        }
        return root;

    }
```





[1 5 3 4 6]

Class BSTPair {
Long max = Long.minval
Long min = Long.maxval
boolean isBST = True
}

```java
public BstPair ValidBST(TreeNode root) {
        if (root == null) {
            return new BstPair();
        }

        BstPair lbp = ValidBST(root.left);
        BstPair rbp = ValidBST(root.right);
        BstPair sbp = new BstPair();
        sbp.min = Math.min(root.val, Math.min(lbp.min, rbp.min));
        sbp.max = Math.max(root.val, Math.max(lbp.max, rbp.max));
        sbp.isbst = lbp.isbst && rbp.isbst && lbp.max < root.val && rbp.min > root.val;
    }
```

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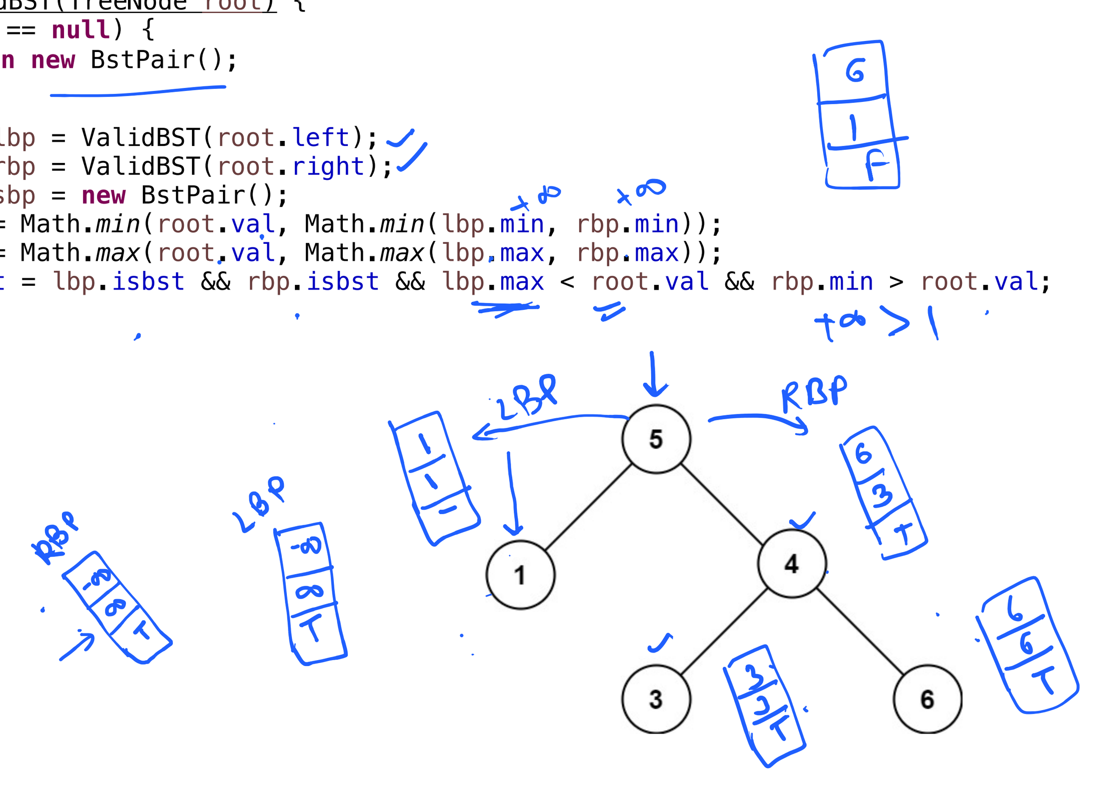

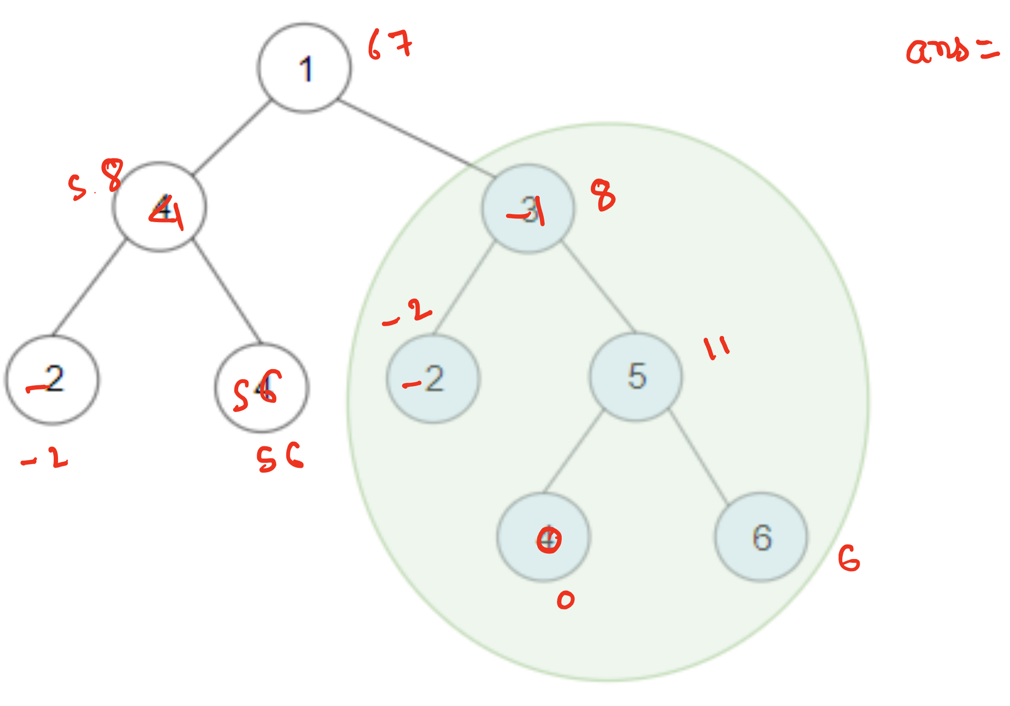

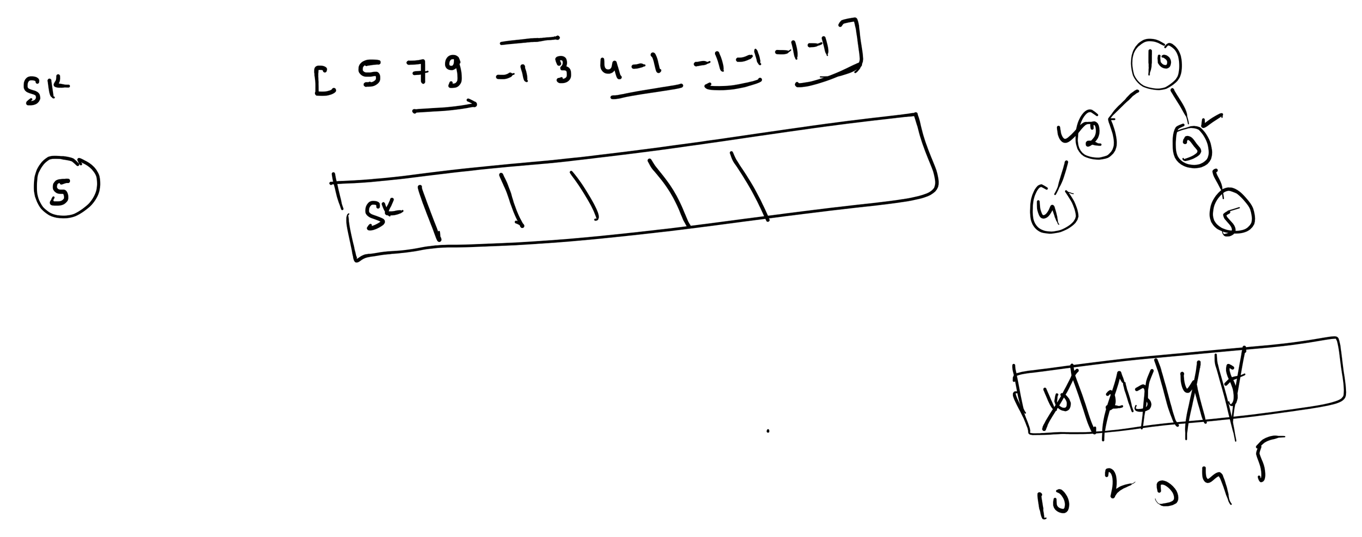


```java
private void createTree() {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int v = sc.nextInt();
        root = new Node(v);
        Queue<Node> q = new LinkedList<>();
        q.add(root);
        while(!q.isEmpty()) {
            Node r =q.poll();
            int c1=sc.nextInt();
            int c2=sc.nextInt();
            if(c1!=-1) {
                r.left=new Node(c1);
                q.add(r.left);
            }
            if(c2!=-1) {
                r.right=new Node(c2);
                q.add(r.right);
            }

        }
    }
```